Database Systems: Course Overview

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Instructor

- Sang-Wook Kim (Big Data Science Lab.)
  - Areas of interest
    - Big data, machine learning, data mining, databases, recommender systems, and social network analysis
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Sang-Wook Kim

- Professional Experiences
  - Professor, Computer Science Department, Hanyang University (2003-Present)
  - Distinguished Professor, Hanyang University, (2024-Present)
  - Director, Institute of Advanced Artificial Intelligence, (2025-Present)
  - Director, BK21PLUS/BK21FOUR Research Program (2013-Present)
  - Non-executive Director, NHN (2019-2024)
  - Member, National Academy Engineering of Korea (2019-Present)
  - Advisory Board Member, Samsung Electronics (2011-2012)
  - Visiting Scientist, Carnegie Mellon University (2009-2011)
  - Post-Doc., IBM Watson Research Center (2000-2001)
- Research Areas
  - Databases, machine learning, data science
- Publications
  - More than 150 international journal papers
  - More than 250 international conference papers
  - More than 150 patents

Sang-Wook Kim

- Professional Activities
  - President, Database Society of Korea (2017-2018)
  - Associate Editor, Information Sciences (2016-Present)
  - Editor, Computer Science and Information Systems (ComSIS), (2017-Present)
  - Editor, Journal of Information and Data Management (JIDM), (2015-Present)
  - General Chairs, WIMS, 2018; CCCI, 2014
  - Program Chairs, ACM IMCOM, 2017; Intelligent Environment (IE), 2017
- Awards
  - Korea Presidential Award – National R&D (2017)
  - Top 100 Best National R&D Projects (2107) – Ranked 1st in CS&EE field
  - Outstanding Research at Hanyang (2016)
  - Best BK21PLUS Research Team (2016)
  - Best Paper Award, IEEE IC-NIDC (2016)
  - Best Paper Award, ACM SAC (2014)
  - Best Poster Award, ACM CIKM (2013)

Goals

- DBMS (database management systems)
  - Basic concepts
  - System architecture
- Database design
  - Entity-relationship model
  - Relational model
- SQL (structured query language)
- Application development
- System components
  - Storage systems
  - Query processing
  - Concurrency control and recovery

Requirements

- C++, Java or **Python** programming skills (mandatory!)
- Data structures (mandatory!)
- File structures

Text Information

- Primary textbook
  - Fundamentals of Database Systems
    - Authors: Elmasri and Navathe
    - Publisher: Benjamin Cummings
- Secondary textbooks
  - Korth and Silbershatz's
  - Ullman and Widom's

Grading Scheme

- Relative evaluation
  - **A:B:CDF = 25%:30%:45%**
- Weights on graded parts
  - Midterm exam: **35 %**
  - Final exam: **35 %**
  - Programing projects: **25 %**
  - Participation : **5 %**
    - **Bonus for participation (good questions or answers)**

Notice – Flipped Learning

- Lecture (online recorded video)
  - The video will be uploaded on **every Tuesday**
  - Lecture video will be provided **during only one week**
    - Video will be uploaded at the class time and closed one week later
    - Watch the lecture **WITHIN** this period (i.e., not after nor before the period) for the attendance to be counted correctly
    - It will be no longer available after the period
- Practice class (offline)
  - The class will be held offline on **every Thursday** (ITBT 203)
  - Sometimes, the professor could join to summarize the lecture and to have Q&A sessions (Once or twice a month)

Programming Projects

- Programing projects: 25% (B+ tree:DB = 3:7)
  - B+ tree implementation (C++, Java or Python)
    - Due: 24 Sept. 23:59 (NOTE!)
  - Database application development using JDBC (Java or Python)
    - Step 1: Requirement analysis (20%)
    - Step 2: Conceptual design (20%)
    - Step 3: Logical design (20%)
    - Step 4: Database construction & application development (40%)

Notice

- Grade ‘D‘ will be given if any two of programing projects are *not successfully fulfilled*
- Grade ‘F‘ will be given if
  - S/he copies somebody else‘s program (i.e., from classmate or ***from the Internet***) or ***allows others to copy her/his own program;***
  - S/he does not take either the midterm or final exam

- The students who took this class earlier (재수강) will be *down-graded*: A+ => B+; A0 => B0

Notice

- Attendance policy
  - No penalty up to 2 absence (after this point, penalized)
  - Two late attendance will be considered as one absence
  - **Note: the attendance later than 10 minutes after the start time will be regarded as ABSENCE rather than late attendance**
- No (audio/video) recording policy
  - Otherwise, this will be penalized significantly
- Do preview on our textbook for 5 minutes
- Visit our HY-IN community site *at least once a week*
  - *For important announcement (possibly) given*

Exam (Offline!)

- Time
  - Midterm exam: One workday in **10/22** (Thursday)
  - Final exam: One workday in **12/17** (Thursday)
    - If you have a problem with this schedule, please contact me via email by 6 Sept. (Sunday)
    - Otherwise, this schedule will be finalized
- Place
  - ITBT 203

Thank You !

